

Machine Learning for Mechanical Engineers at CoEP Session 8: Linear Regression

Session 8: LinRegr

Linear Regression

Want to know future? Predictions

- How does sales volume change with changes in price.
- How is this affected by changes in the weather?
- How does the amount of a drug absorbed vary with dosage and with body weight of patient?
- Does it depend on blood pressure?

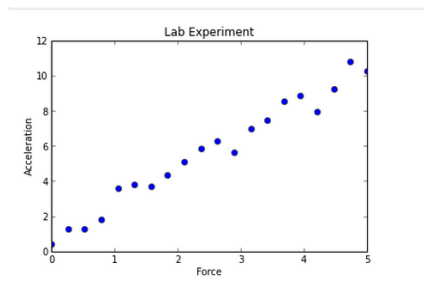
Answering questions like these, requires us to create a model.

The model

- A model is a formula where one variable (response) varies depending on one or more independent variables (covariates).
- The formula (model) can be simple or complex.
- Simplest is a Linear Model where the dependent varies linearly with the independent variable(s).
- Creating a Linear Model involves a technique known as Linear Regression.

Remember? - High School Physics Lab

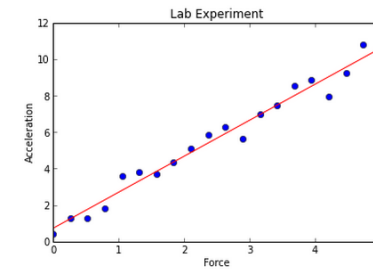
- Finding how acceleration changes when Force is applied
- For some input X (say force), got output Y (say acceleration).
- Plotted the pairs of observations x, y.



What next?

Fitting line

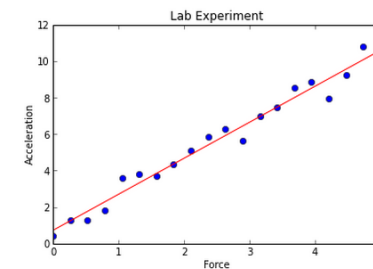
- Then you had to fit a straight line through points
- A visual “best fit”, some points above, some below.
- Found 'm' the slope (how?), and 'b' (how?).



Whats the model here?

Fitting line

- Equation of line, the formula, itself is the model.
- You were doing informal Linear Regression.



Linear Regression

- What: for predicting a quantitative variable
- Age: “it’s been around for a long time”. The term regression was devised by Francis Galton in his article Regression Towards Mediocrity in Hereditary Stature in 1886.
- Complexity: easy to understand and use
- Popularity: still widely used

Possible Techniques

- Statisticians know Regression as OLS (Ordinary Least Square) method to fit a line. Deals with minimization of RMS (Root Mean Square) error.
- Theoretical/Analytical Solution $W = (X^T X)^{-1} X^T y$. Not practical for large matrices. Also, matrix is not invertible if any two rows are same.
- Machine Learning Engineers know Regression as prediction of continuous variables (floats). It uses Gradient Descent for the minimization part.

Linear Regression - Analytical Approach (OLS)

Example: Tip Amount

- In a restaurant, tip you pay to the waiter,
- Typically depends on the bill amount.
- Smaller the bill, lesser the tip, and vice versa.

Can we predict the tip amount just by looking at the bill amount?

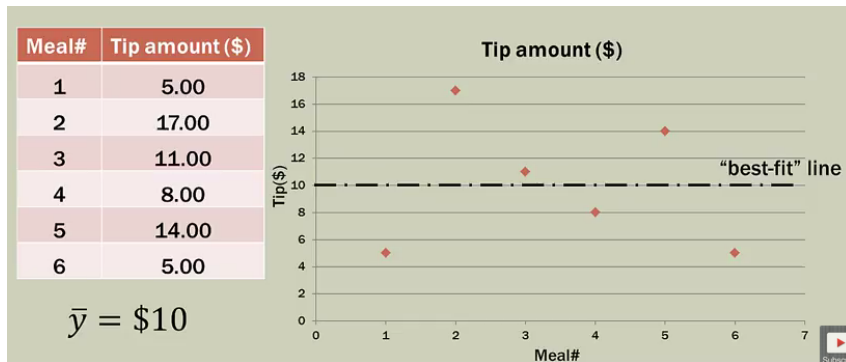
Inputs?

Outputs?

Example: Tip Amount

- You collect, random (meaning, any) 6 tips.
- YOU HAVE NOT COLLECTED THE BILL AMOUNT.
- Just one variable (Meal id is just the descriptor its not a independent variable)
- Any guesses?

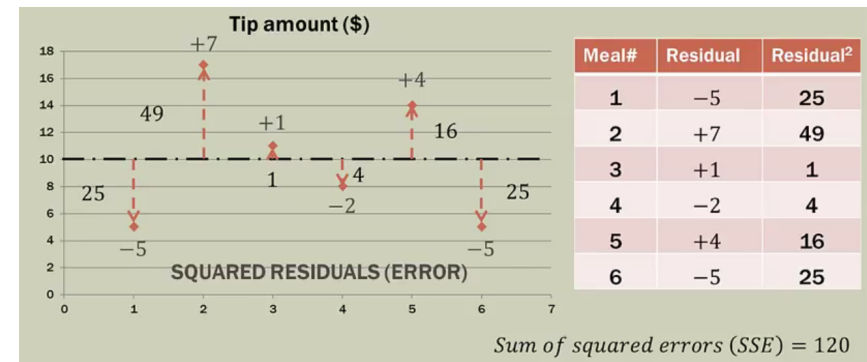
Most Basic Situation



- With just one variable, mean is the best we can do.
- Note: not a single point falls on it.
- Its just an estimate.
- Goodness of fit?

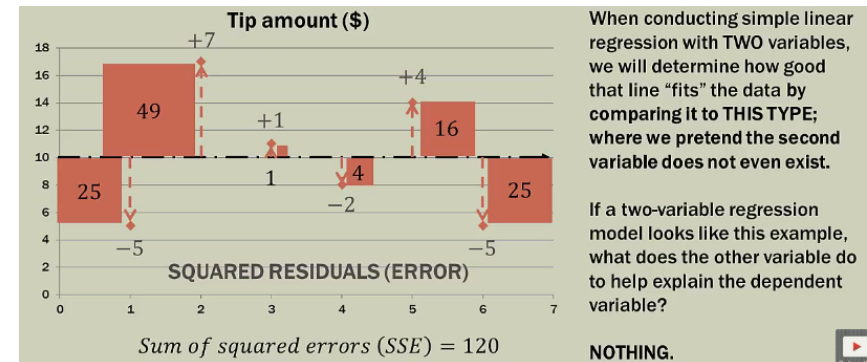
(Reference: Simple Linear Regression: Step-By-Step - Dan Wellisch)

Example: Tip Amount



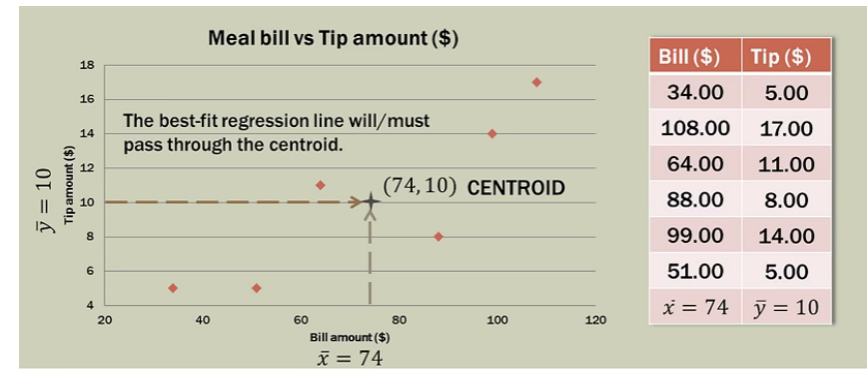
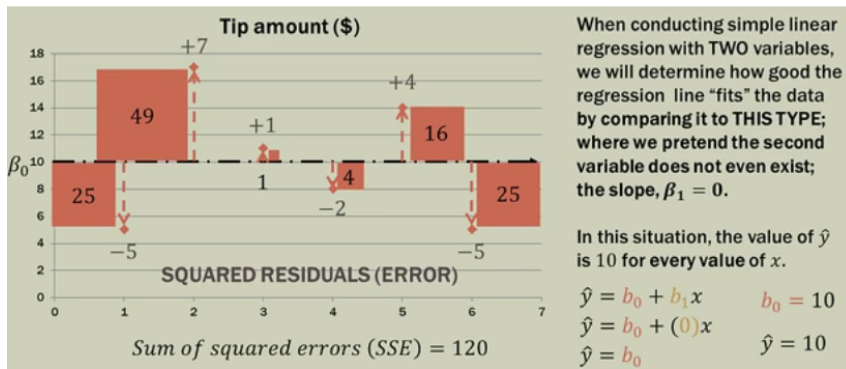
- Std Variation says about goodness of fit.
- Diff of each point with mean are called 'residuals' or 'errors'
- Idea is to minimize SSE (Sum of Squared Errors) or RSS (Residual Sum of Squares)

Important Concept



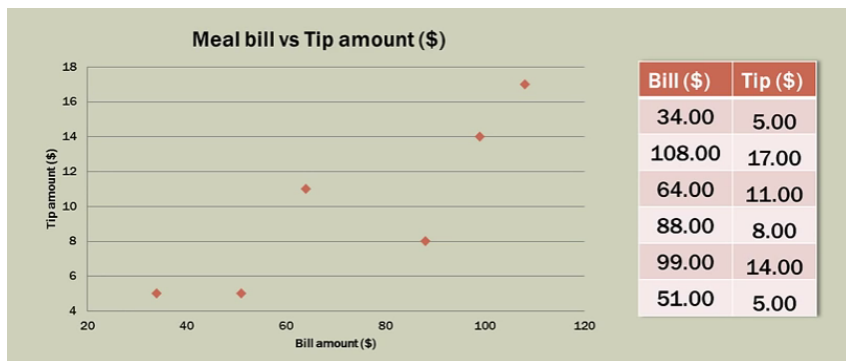
- With just the target variable (tips) we got SSE as 120.
- If we add one more variable to the problem, ie bill amount, what do you expect of SSE?
- It should reduce, or else, whats the use of adding this variable.
- So, we will compare our Linear regression fit line with this (one variable) line.

Important Concept



- We find slope and intercept as b_1 and b_0 respectively.
- A line with $b_1 = 0$ is the horizontal line.
- Its SSE becomes the benchmark. Any selected pair of b_1 and b_0 should give SSE less than the benchmark SSE.

Linear Regression with Two Variables

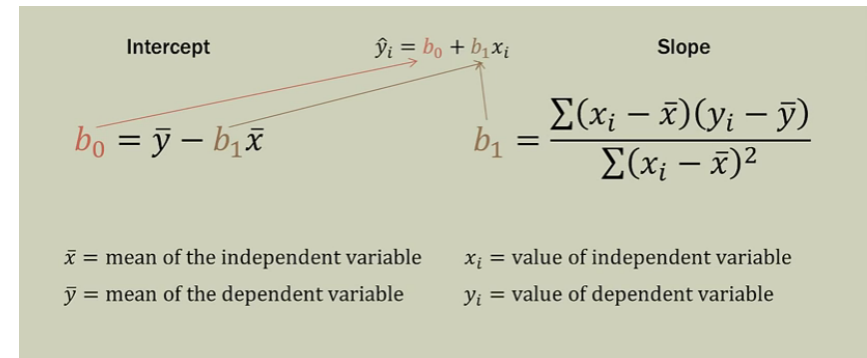


- You can fit a line, however you want.
- Even horizontal line can be put
- But then, which one is the best?
- Need to be at least better than horizontal line!!

Methodology

Hypothesis: A line passing through the centroid of the data points will be good. Lets check.

Calculations



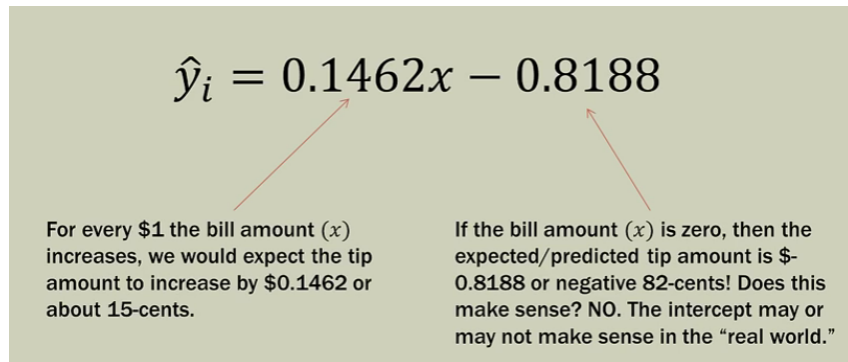
- Intercept and Slope are calculated as above.
- A simple tabular calculations can be done.

Calculations

Meal	Total bill (\$)	Tip amount (\$)	Bill deviation	Tip Deviations	Deviation Products	Bill Deviations Squared
	x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	34	5	-40	-5	200	1600
2	108	17	34	7	238	1156
3	64	11	-10	1	-10	100
4	88	8	14	-2	-28	196
5	99	14	25	4	100	625
6	51	5	-23	-5	115	529
	$\bar{x} = 74$	$\bar{y} = 10$			$\sum = 615$	$\sum = 4206$

- Slope comes to $= 615/4206 = 0.1462$
- Intercept comes to $= 10 - 0.1462(74) = -0.818$
- So, eq is $y = 0.1462x - 0.8188$

Interpretation



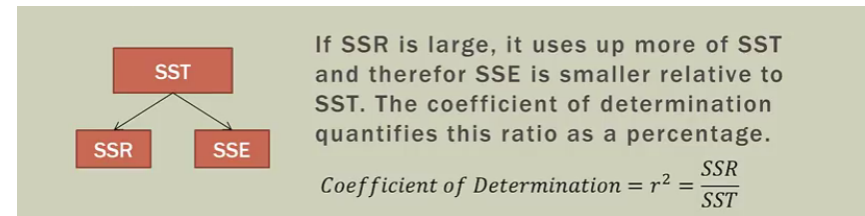
- Slope gives proportionality with which tip is calculated.
- Intercept may or may not make sense in real life.
- Is this manual calculation giving good line?

Evaluation

Meal	Total bill (\$)	Observed tip amount (\$)	$\hat{y}_i$ (predicted tip amount)	Error ($y - \hat{y}_i$)	Squared Error ($(y - \hat{y}_i)^2$)
	x	y			
1	34	5	4.1505	0.8495	0.7217
2	108	17	14.9693	2.0307	4.1237
3	64	11	8.5365	2.4635	6.0688
4	88	8	12.0453	-4.0453	16.3645
5	99	14	13.6535	0.3465	0.1201
6	51	5	6.6359	-1.6359	2.6762
	$\bar{x} = 74$	$\bar{y} = 10$			SSE = $\sum = 30.075$

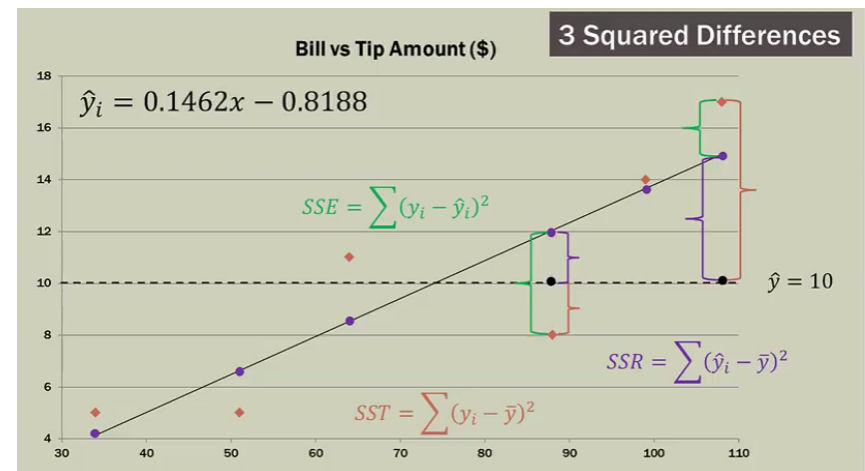
- Sum of Squares of Error, SSE is 30 for our manual model
- Its far less than the horizontal line benchmark of 120.
- So, by adding Bill amount as independent variable, reduced error by about 90.

Evaluation



- SST (sum of squares total) is the benchmark error. Its constant is 120.
- SSE (sum of squares error) is the level to which we could get error to it by applying regression is 30.
- The difference, the achievement by regression is called SSR (sum of squares by regression) is 90.

Evaluation



- For each x , there is predicted y and actual y
- 3 relationships: observed, predicted and total
- Can decide quality based on the ratios among them.

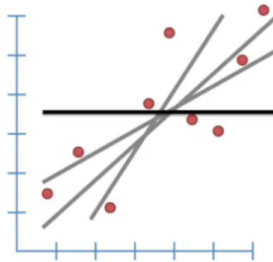
R^2 Statistics

- $SST = SSR + SSE$
- SST : Total variance in the response Y . Amount of variability inherent in the response, before the regression is performed. y_i to $\bar{y}$
- It has two components, one explained by this regression model, other the unexplained.
- SSR : The amount of variability that is explained after performing the regression. $\hat{y}_i$ to $\bar{y}$
- SSE : The amount of variability that is left unexplained, that our model could not minimize. y_i to $\hat{y}_i$

Linear Regression - Machine Learning Approach

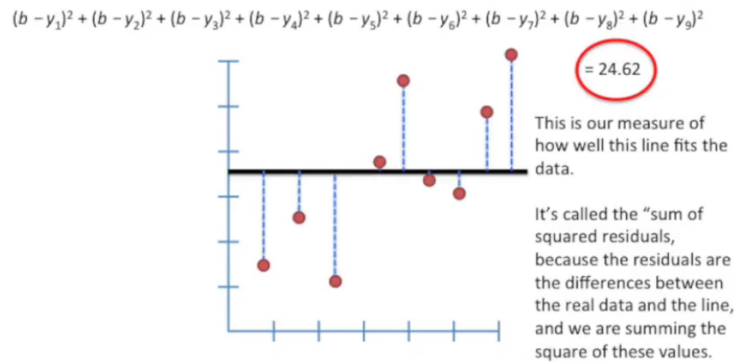
Linear regression

- We have multiple observations of inputs and an output
- Object: to find a model, ie a formulation, adhering to more or less all data points
- ie. to attempt to find the “best” line
- But, which line is the “best”?
- The horizontal line is the worst, the benchmark. We need to beat that!!



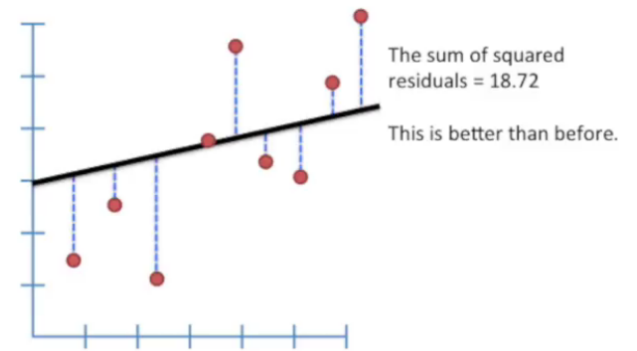
(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Benchmark Line



(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Rotated Line



Its better but is this best?

(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Rotated Line

Rotated a bit more. Better?

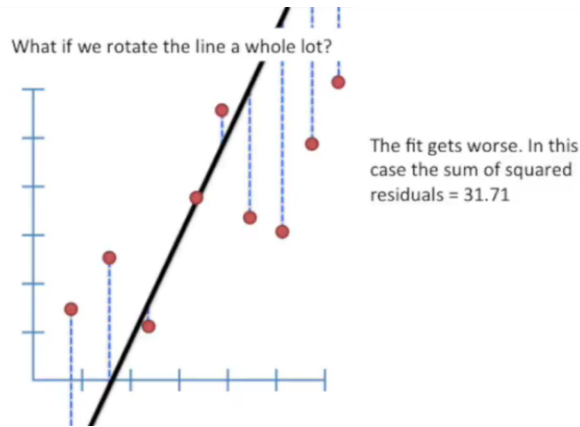


Its better but is this best?

(Ref: StatQuest: Linear Models - Josh Starmer)

Sum of Squared Errors for Rotated Line

Why not rotate a whole lot!!

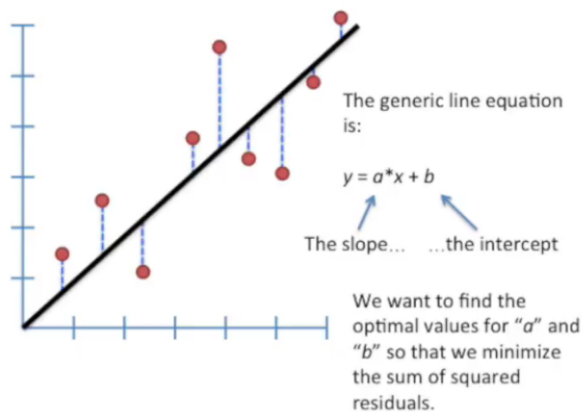


No, we lost the BEST point somewhere in between. What's that "optimized" point?

(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization

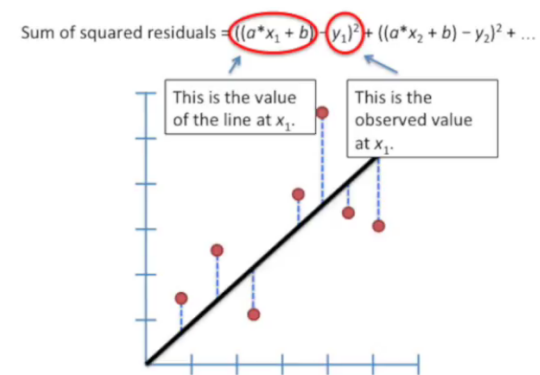
To find the sweet spot is the optimized point, let's put some formulation based on generic line equation.



(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization

Mathematically ...

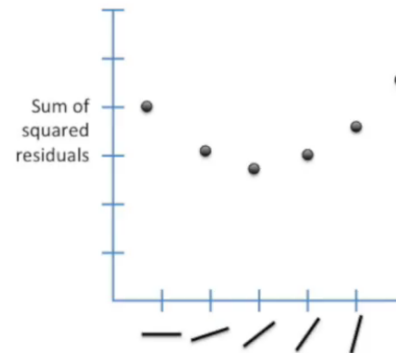


Since we want the line that will give us the smallest sum of the squares of errors (SSE: Sum of Squared Errors), this method for finding the best values for "a" and "b" is called as "Least Squares".

(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization

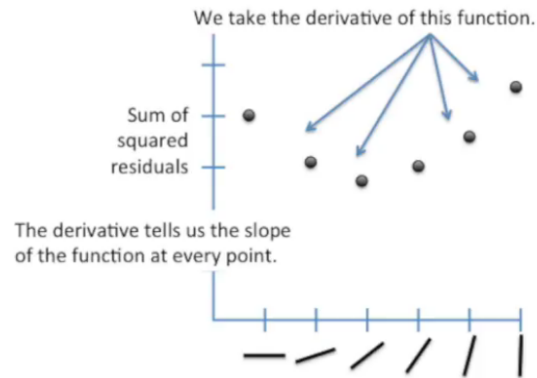
- If we plotted SSE for each rotated line the plot looks like.
- We can see that SSE was high for the horizontal line, then came down as we started rotating the line,
- But started rising again, if we passed the sweet spot.



(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization

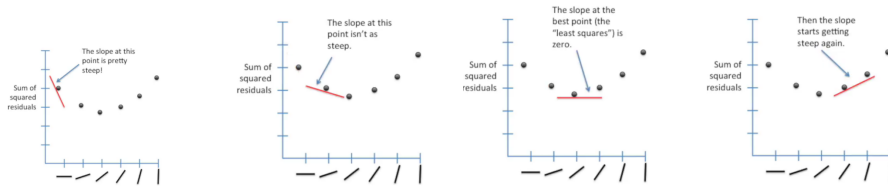
- We need to get to the bottom point.
- We can start at any (random) point, and SLIDE down to bottom.
- SLIDING means going along slope.
- Slope is given by Derivative of the (SSE) function/curve, at the point



(Ref: StatQuest: Linear Models - Josh Starmer)

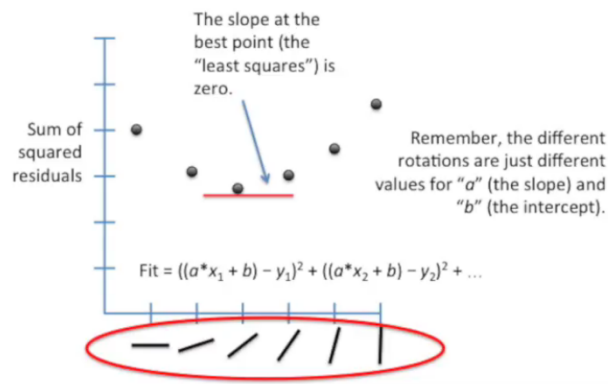
Optimization

- Slope starts decreasing from end, faster, then slower,
- Reaches 0, horizontal, and then starts increasing again.
- Need to find the “0” slope point.



These different slopes means different “a” parameters (Ref: StatQuest: Linear Models - Josh Starmer)

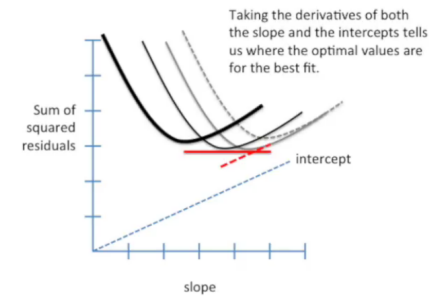
Optimization



(Ref: StatQuest: Linear Models - Josh Starmer)

Optimization

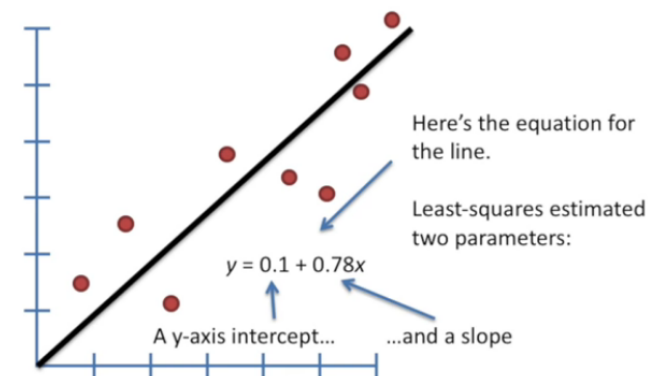
- For “a” and “b” both to be optimized, we will need 3D picture, as the Loss function would then take Z axis
- It will be a Surface.
- Derivative will be Partial, but the objective is the same, Bottom-most point. ie Minimum Loss function value



The “a” and “b” coordinates corresponding to the bottom-most point give most minimum Loss. So it represents the “best” line.

(Ref: StatQuest: Linear Models - Josh Starmer)

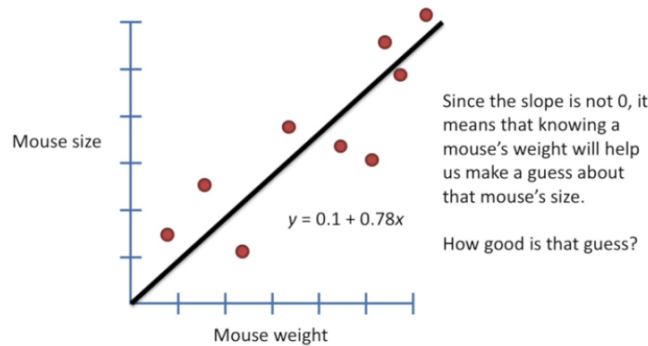
Results



(Ref: StatQuest: Linear Models - Josh Starmer)

Example: Mouse Size

- We have data points having mouse size and mouse weight
- We wish to find mouse size given mouse weight.



we use R^2 to find “good-ness” of fit (of line)
(Ref: StatQuest: Linear Models - Josh Starmer)

Linear regression For Complex Problems

- In real life, problems are far more complex than just one x and one y .
- There could be many x 's i.e. many inputs which decide the output
- General formulation : Predicting quantitative response y based on predictor variables x . Assumes linear relationship between x and y

$$y_i = x_{1,i}\beta_1 + x_{2,i}\beta_2 + \dots + x_{p,i}\beta_p + \epsilon_i$$

- Objective is to find the parameters β_j that minimize the squared residuals
- There is no slope formula for these β s. So, we will solve machine learning way.

Linear Regression - Advertising Budget Example

Example: Advertising Dataset

- Toy Dataset: Advertising, Advertising.csv
- Sales totals for a product in 200 different markets. Advertising budget in each market, broken down into TV, radio, newspaper

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	9.3
3	151.5	41.3	58.5	18.5
4	180.8	10.8	58.4	12.9

(Data: <http://www-bcf.usc.edu/~gareth/ISL/data.html>)

Problem

- Goal: What marketing plan for next year will result in high product sales?
- Questions (Hypothesis):
 - Is there a relationship between advertising budgets and sales?
 - How strong is the relationship between advertising budgets and sales?
 - Which media contribute to the sales the most?
 - Is the relationship linear?

EDA

Correlation Matrix

	TV	Radio	Newspaper	Sales
TV	1.000000	0.054809	0.056648	0.782224
Radio	0.054809	1.000000	0.354104	0.576223
Newspaper	0.056648	0.354104	1.000000	0.228299
Sales	0.782224	0.576223	0.228299	1.000000

- Correlation between radio and newspaper is 0.35
- Barely any correlation (or “not correlated”) for TV/radio and TV/newspaper

EDA

	TV	Radio	Newspaper	Sales
TV	1.000000	0.054809	0.056648	0.782224
Radio	0.054809	1.000000	0.354104	0.576223
Newspaper	0.056648	0.354104	1.000000	0.228299
Sales	0.782224	0.576223	0.228299	1.000000

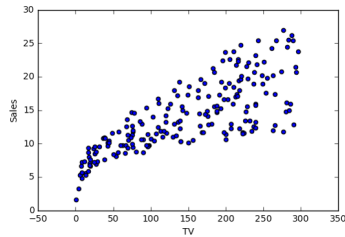
- Reveals tendency to spend more on Newspaper advertising in markets where more is spent on Radio advertising.
- Sales higher in markets where more is spent on Radio, but more also tends to be spent on Newspaper.
- In Simple LM: Newspaper “gets credit” for effect of Radio on Sales.

Advertising Dataset

- What are some ways we can regress sales onto advertising using Simple Linear Regression?
- One model:

$$\begin{aligned} \text{sales} &\approx \beta_0 + \beta_1 \times TV \\ y &\approx \beta_0 + \beta_1 \times x \end{aligned}$$

- Scatter plot visualization for TV and Sales.

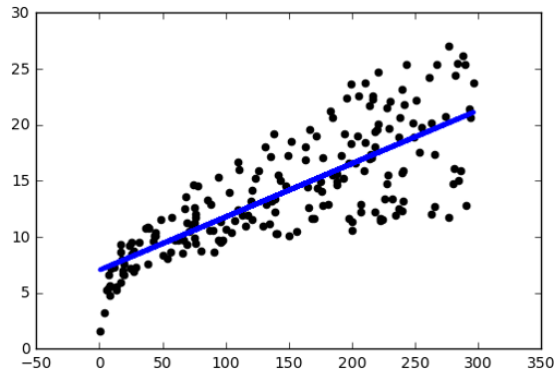


Advertising Dataset

- Simple Linear Model in Python (using pandas and scikit): Predictor: x , Response: y
- $\text{Sales} = 7.03259 + 0.04754 \times TV$

Advertising Dataset

Scatter plot visualization for TV and Sales with Linear Model.



If we have 3 such models, can we “somehow” compose them (say, averaging) to get the combined model?

Multiple Linear Regression

- In practice, often have more than one predictor
 - $\text{sales} \approx \beta_{0,1} + \beta_{1,1} \times TV$
 - $\text{sales} \approx \beta_{0,2} + \beta_{1,2} \times \text{Newspaper}$
 - $\text{sales} \approx \beta_{0,3} + \beta_{1,3} \times \text{radio}$
- Run three separate simple linear regressions
- For p predictor variables: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$
- Since error ϵ has mean zero, we usually omit it.
- A one-unit change in any predictor variable x_i will change the expected mean response by β_j units. $\text{sales} = \beta_0 + \beta_1 \times TV + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper}$

Estimating the Parameters $\beta_0, \beta_1, \dots$

- Advertising Dataset $\text{sales} = \beta_0 + \beta_1 \times TV + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper}$
- After solving $\text{sales} = 2.93 + 0.0457 \times TV + 0.188 \times \text{radio} + -0.0010 \times \text{newspaper}$

Coefficients

- Many Simple Linear Regression
 - $TVModel : [[0.04753664]][7.03259355]$
 - $RadioModel : [[0.20249578]][9.3116381]$
 - $NewspaperModel : [[0.0546931]][12.35140707]$
- Single Multiple Linear Regression
 - $Coefficients : [[0.04576465, 0.18853002, -0.00103749]]$
 - $Intercept : [2.93888937]$

Answering questions using Regression Model

Goal: What marketing plan for next year will result in high product sales?

- Question: Is there a relationship between advertising budget and sales?
- Answer: Yes, hypothesis testing shows that we can reject the null hypothesis that $B_{TV} = B_{RADIO} = B_{NEWSPAPER} = 0$

In conclusion

- Pros of Linear Regression Model:
 - Provides nice interpret-able results
 - Works well on many real-world problems
- Cons of Linear Regression Model:
 - Assumes linear relationship between response and predictors: Change in the response Y due to a one-unit change in X_i is constant
 - Assumes additive relationship. Effect of changes in a predictor X_i on response Y is independent of the values of the other predictors